1. O/N/18/21/Q6

Fig. 6.1 is the view from above of a stationary magnet and a small compass on a laboratory bench.

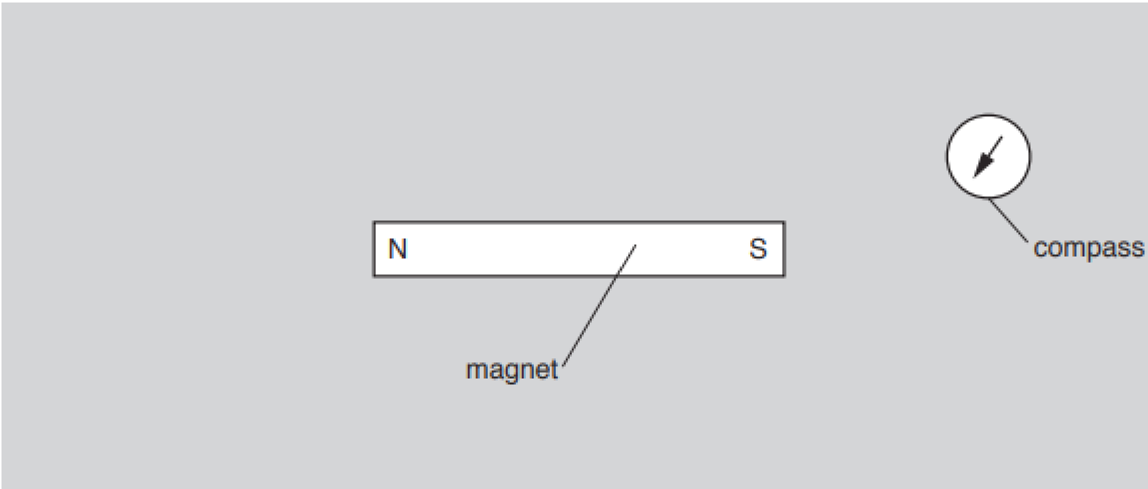

Fig. 6.1

(a) State the material from which the compass needle is made.

.... [1]

(b) Describe how the small compass is used to plot magnetic field lines in the region surrounding the magnet.

....

....

....

....

.... [3]

(c) An unmagnetised iron bar PQ is placed near to the magnet on the laboratory bench.

Fig. 6.2 shows the two poles of the magnet and PQ.

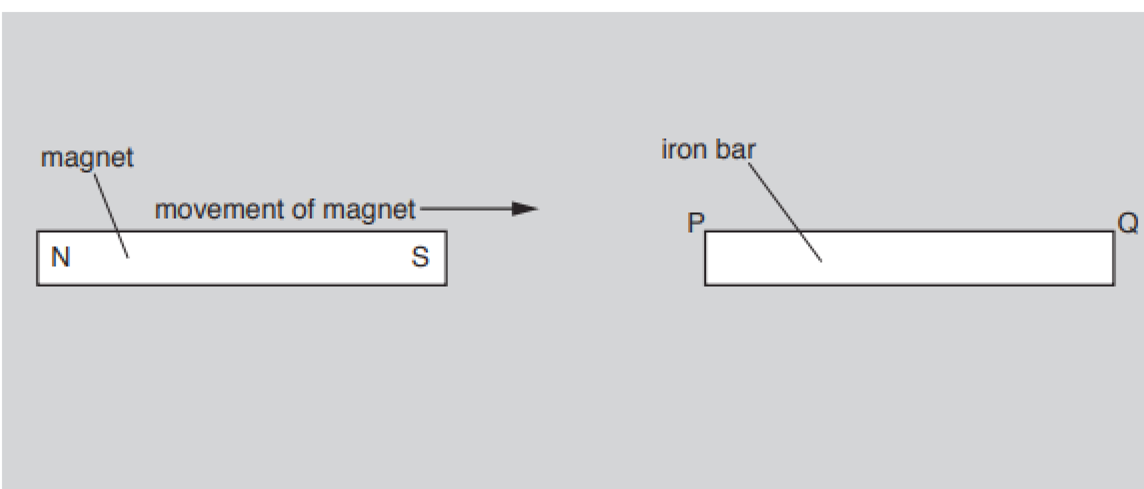

Fig. 6.2

The iron bar is initially at rest and the magnet is moved to the right very slowly.

When the magnet is a short distance from the iron bar, the iron bar moves very quickly to the left towards it.

Explain why the iron bar is attracted to the magnet.

....

....

....

.... [3]

2. M/J/14/22/Q7

Fig. 7.1 shows a compass needle near a bar magnet. Magnetic poles are shown on the compass needle and on the magnet.

A finger stops the compass needle from turning.

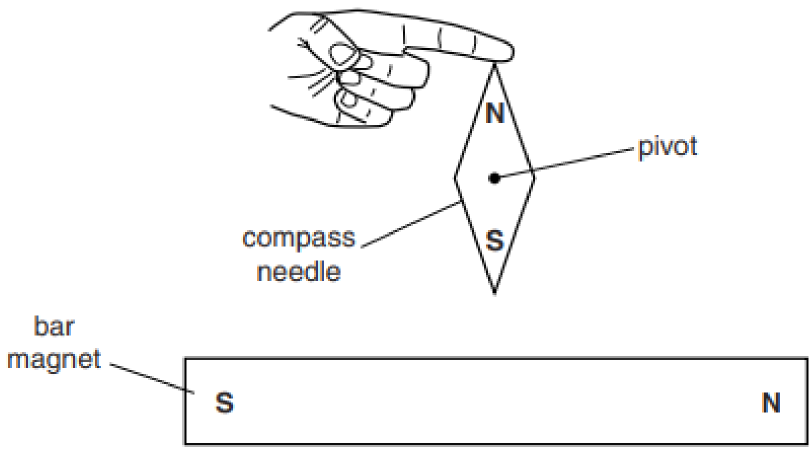

Fig. 7.1 (not to scale)

(a) **(i)** The magnet causes a force on the S-pole of the compass needle.

On Fig. 7.1, draw an arrow from the S-pole of the compass needle to show the direction of this force. [1]

(ii) Explain why the compass needle turns when the finger is removed.

....

.... [1]

(b) A small compass is used to plot the magnetic field lines of the magnet.

Describe how the compass is used to plot magnetic field lines on a piece of paper.

....

....

....

....

.... [3]

3. O/N/13/21/Q6

(a) Suggest a material that is used to make a permanent magnet.

.... [1]

(b) When there are no other magnetic fields present, the needle of a plotting compass points due north in the Earth's magnetic field. This is shown in Fig. 6.1.

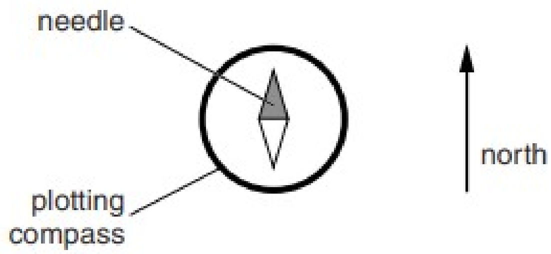

Fig. 6.1

The plotting compass is placed close to a permanent magnet, first at position A and then at position B, as shown in Fig. 6.2.

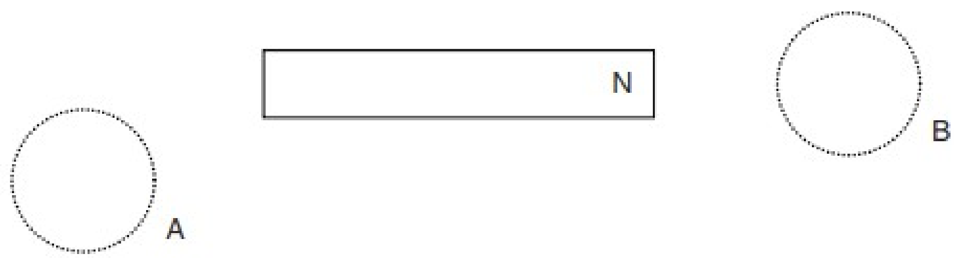

Fig. 6.2

The magnetic field close to the permanent magnet is very much larger than the magnetic field of the Earth.

On Fig. 6.2, draw needles in the two circles to show the direction in which the compass needle points when the compass is at A and at B. [2]